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Method and system for generating fueling instructions for a vehicle

Abstract

A system (900) and method (1100) instructing a vehicle (910) where and when to refuel is disclosed herein. The system (900) comprises a server (925) for receiving a workflow for a vehicle (910). The workflow comprises an origination location (901) of the vehicle, a destination (950) of the vehicle (910), a route (905) to the destination, a cargo, a time of departure and a time of arrival. The server (925) determines a plurality of fuel stops (930) along the route (905).

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Background/Summary

CROSS REFERENCES TO RELATED APPLICATIONS (1) The Present Application is a continuation application of U.S. patent application Ser. No. 18/126,458, filed on Mach 26, 2023, which is a continuation application of U.S. patent application Ser. No. 17/536,066, filed on Nov. 28, 2021, now U.S. Pat. No. 11,641,677, issued on May 2, 2023, which is a continuation application of U.S. patent application Ser. No. 16/912,265, filed on Jun. 25, 2020, now U.S. Pat.

No. 11,197,329, issued on Dec. 7, 2021, which claims priority to U.S. Provisional Patent Application No. 62/867,845, filed on Jun. 27, 2019, and U.S. patent application Ser. No. 16/912,265 is also a continuation-in-part application of U.S. patent application Ser. No. 16/870,955, filed on May 9, 2020, now U.S. Pat. No. 11,330,644, issued on May 10, 2022, which is a continuation-in-part application of U.S. patent application Ser. No. 16/416,396, filed on May 20, 2019, now U.S. Pat. No. 10,652,935, issued on May 12, 2020, which is a continuation-in-part application of U.S. patent application Ser. No. 16/118,436, filed on Aug. 31, 2018, now U.S. Pat. No. 10,334,638, issued on Jun. 25, 2019, which is a continuation application of U.S. patent application Ser. No. 15/917,633, filed on Mar. 11, 2018, now U.S. Pat. No. 10,070,471, issued on Sep. 4, 2018, which is a continuation application of U.S. patent application Ser. No. 15/624,814, filed on Jun. 16, 2017, now U.S. Pat. No. 9,961,710, issued on May 1, 2018, which claims priority to U.S. Provisional Patent Application No. 62/352,014, filed on Jun. 19, 2016, and U.S. patent application Ser. No. 16/912,265 is a continuation-in-part application of U.S. patent application Ser. No. 16/664,906, filed on Oct. 27, 2019, now U.S. Pat. No. 10,803,682, issued on Oct. 13, 2020, which is a continuation application of U.S. patent application Ser. No. 15/859,380, filed on Dec. 30, 2017, now U.S. Pat. No. 10,475,258, issued on Nov. 12, 2019, which is a continuation-in-part application of U.S. patent application Ser. No. 15/624,814, filed Jun. 16, 2017, now U.S. Pat. No. 9,961,710, issued on May 1, 2018, which claims priority to U.S. Provisional Patent Application No. 62/352,014, filed on Jun. 19, 2016, and U.S. patent application Ser. No. 15/859,380 claims priority to U.S. Provisional Patent Application No. 62/441,290, filed on Dec. 31, 2016, U.S. Provisional Patent Application No. 62/441,298, filed on Dec. 31, 2016, and U.S. Provisional Patent Application No. 62/441,315, filed on Dec. 31, 2016, each of which is hereby incorporated by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) Not Applicable

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention generally relates to remote instructions for a vehicle.

Description of the Related Art

(3) The prior art discusses various techniques for wireless networks for vehicles.

(4) U.S. Pat. No. 9,215,590 for Authentication Using Vehicle Data Pairing discloses the wireless pairing of a portable device with an on-board computer of a vehicle for authenticating a transaction with a third party.

(5) General definitions for terms utilized in the pertinent art are set forth below.

(6) Beacon is a management frame that contains all of the information about a network. In a WLAN, Beacon frames are periodically transmitted to announce the presence of the network.

(7) BLUETOOTH technology is a standard short range radio link that operates in the unlicensed 2.4 gigaHertz band.

(8) FTP or File Transfer Protocol is a protocol for moving files over the Internet from one computer to another.

(9) Hypertext Transfer Protocol (“HTTP”) is a set of conventions for controlling the transfer of information via the Internet from a web server computer to a client computer, and also from a client computer to a web server, and Hypertext Transfer Protocol Secure (“HTTPS”) is a communications protocol for secure communication via a network from a web server computer to a client computer, and also from a client computer to a web server by at a minimum verifying the authenticity of a web site.

(10) Internet is the worldwide, decentralized totality of server computers and data-transmission paths which can supply information to a connected and browser-equipped client computer, and can receive and forward information entered from the client computer.

- (11) Media Access Control (MAC) Address is a unique identifier assigned to the network interface by the manufacturer.
- (12) Memory generally includes any type of integrated circuit or storage device configured for storing digital data including without limitation ROM, PROM, EEPROM, DRAM, SDRAM, SRAM, flash memory, and the like.
- (13) Organizationally Unique Identifier (OUI) is a 24-bit number that uniquely identifies a vendor, manufacturer, or organization on a worldwide basis. The OUI is used to help distinguish both physical devices and software, such as a network protocol, that belong to one entity from those that belong to another.
- (14) Processor generally includes all types of processors including without limitation microprocessors, general purpose processors, gate arrays, array processors, application specific integrated circuits (ASICs) and digital signal processors.
- (15) SCP (Secure Connection Packet) is used to provide authentication between multiple devices or a local party and remote host to allow for secure communication or the transfer of computer files.
- (16) SSID (Service Set Identifier) is a 1 to 32 byte string that uniquely names a wireless local area network.
- (17) Transfer Control Protocol/Internet Protocol ("TCP/IP") is a protocol for moving files over the Internet.
- (18) URL or Uniform Resource Locator is an address on the World Wide Web.
- (19) User Interface or UI is the junction between a user and a computer program. An interface is a set of commands or menus through which a user communicates with a program. A command driven interface is one in which the user enter commands. A menu-driven interface is one in which the user selects command choices from various menus displayed on the screen.
- (20) Web-Server is a computer able to simultaneously manage many Internet information-exchange processes at the same time. Normally, server computers are more powerful than client computers, and are administratively and/or geographically centralized. An interactive-form information-collection process generally is controlled from a server computer, to which the sponsor of the process has access.
- (21) There is a need for informing a vehicle such as a truck, where and when to refuel during a delivery.

BRIEF SUMMARY OF THE INVENTION

- (22) The present invention provides a system and method instructing a vehicle where and when to refuel.
- (23) One aspect of the present invention is a method for instructing a vehicle where and when to refuel. The method includes receiving at a server a workflow for a vehicle. The workflow comprises an origination location of the vehicle, a destination of the vehicle, a route to the destination, a cargo, a time of departure and a time of arrival. The method also includes receiving at the server a real-time driver profile for the driver of the vehicle. The method also includes receiving at the server real-time data for the vehicle from a CVD connected to on board diagnostics for the vehicle. The method also includes receiving at the server a configuration of the vehicle. The method also includes receiving at the server a plurality of dynamic compliance rules. The method also includes receiving at the server a plurality of configurable real-time vehicle data trigger events. The method also includes determining at the server a real-time GPS location for the vehicle. The method also includes determining at the server a plurality of fuel stops along the route. The method also includes calculating a fuel stop from the plurality of fuel stops for the vehicle based on the workflow, the real-time driver profile, the configuration of the vehicle, the real-time GPS location of the vehicle, the real-time vehicle data, the plurality of dynamic compliance rules, and the selected fuel station profile. The method also includes transmitting to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump of the fuel stop for fueling during a predetermined fueling time

period.

(24) Another aspect of the present invention is a method for instructing a vehicle where and when to refuel. The method includes determining a real-time GPS location for a vehicle. The method also includes determining a plurality of fuel stops along a route for the vehicle. The method also includes calculating a fuel stop from the plurality of fuel stops for the vehicle based on a workflow, a real-time driver profile, a configuration of the vehicle, the real-time GPS location of the vehicle, real-time vehicle data, a plurality of dynamic compliance rules, and a selected fuel stop profile. The method also includes transmitting to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump of the fuel stop for fueling during a predetermined time period.

(25) Yet another aspect of the present invention a non-transitory computer-readable medium that stores a program that causes a processor to perform functions for instructing a vehicle where and when to refuel. The functions include determining a real-time GPS location for a vehicle. The functions also include determining a plurality of fuel stops along a route for the vehicle. The functions also include calculating a fuel stop from the plurality of fuel stops for the vehicle based on a workflow, a real-time driver profile, a configuration of the vehicle, the real-time GPS location of the vehicle, real-time vehicle data, a plurality of dynamic compliance rules, and a selected fuel stop profile. The functions also include transmitting to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump of the fuel stop for fueling during a predetermined time period.

(26) Having briefly described the present invention, the above and further objects, features and advantages thereof will be recognized by those skilled in the pertinent art from the following detailed description of the invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of system for a secure communication protocol for connecting a wireless device to a single access point in a vehicle.
- (2) FIG. 1A is a continuation of the block diagram of FIG. 1.
- (3) FIG. 2 is a flow chart of a method for a secure connection to a wireless network of a vehicle.
- (4) FIG. 3 is an illustration of a driver identifying a vehicle through connection of a tablet computer to an unpublished network.
- (5) FIG. 4 is an isolated view of general electrical components of a mobile communication device.
- (6) FIG. 5 is an isolated view of general electrical components of a server.
- (7) FIG. 6 is a flow chart of method for securely connecting a wireless device to a single access point in a vehicle.
- (8) FIG. 7 is an illustration of a system for securely connecting a wireless device to a single access point in a vehicle.
- (9) FIG. 8 is an illustration of a driver identifying a vehicle through connection of a tablet computer to an unpublished network.
- (10) FIG. 9 is a block diagram of a system for instructing a vehicle where and when to refuel.
- (11) FIG. 10 is an illustration of multiple sensors on a truck.
- (12) FIG. 11 is a flow chart for a method for instructing a vehicle where and when to refuel.
- (13) FIG. 12 is a flow chart for a method for instructing a vehicle where and when to refuel.

DETAILED DESCRIPTION OF THE INVENTION

(14) A first embodiment is a method for instructing a vehicle where and when to refuel. The method includes receiving at a server a workflow for a vehicle. The workflow comprises an origination location of the vehicle, a destination of the vehicle, a route to the destination, a cargo, a

time of departure and a time of arrival. The method also includes receiving at the server a real-time driver profile for the driver of the vehicle. The method also includes receiving at the server real-time data for the vehicle from a CVD connected to on board diagnostics for the vehicle. The method also includes receiving at the server a configuration of the vehicle. The method also includes receiving at the server a plurality of dynamic compliance rules. The method also includes receiving at the server a plurality of configurable real-time vehicle data trigger events. The method also includes determining at the server a real-time GPS location for the vehicle. The method also includes determining at the server a plurality of fuel stops along the route. The method also includes calculating a fuel stop from the plurality of fuel stops for the vehicle based on the workflow, the real-time driver profile, the configuration of the vehicle, the real-time GPS location of the vehicle, the real-time vehicle data, the plurality of dynamic compliance rules, and the selected fuel station profile. The method also includes transmitting to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump of the fuel stop for fueling during a predetermined fueling time period.

(15) A second embodiment is a method for instructing a vehicle where and when to refuel. The method includes determining a real-time GPS location for a vehicle. The method also includes determining at least one fuel stop along a route for the vehicle. The method also includes calculating the distance and time to the fuel stop for the vehicle based on a workflow, a real-time driver profile, a configuration of the vehicle, the real-time GPS location of the vehicle, real-time vehicle data, a plurality of dynamic compliance rules, and a selected fuel stop profile. The method also includes transmitting to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump of the fuel stop for fueling during a predetermined time period.

(16) A third embodiment is a non-transitory computer-readable medium that stores a program that causes a processor to perform functions for instructing a vehicle where and when to refuel. The functions include determining a real-time GPS location for a vehicle. The functions also include determining a plurality of fuel stops along a route for the vehicle. The functions also include calculating a best fuel stop from the plurality of fuel stops for the vehicle based on a workflow, a real-time driver profile, a configuration of the vehicle, the real-time GPS location of the vehicle, real-time vehicle data, a plurality of dynamic compliance rules, and a selected fuel stop profile. The functions also include transmitting to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump of the fuel stop for fueling during a predetermined time period.

(17) The real-time data for the vehicle comprises a real-time speed of the vehicle, tire pressure values from a plurality of tire sensors, refrigeration/HVAC unit values, a plurality of fluid levels, a plurality of power unit values, a real-time fuel tank capacity, and a fuel type.

(18) The plurality of configurable real-time vehicle data trigger events comprises a value outside of a predetermined range for the real-time data of the vehicle.

(19) The method also automatically bills an off-site entity for the fuel pumped by the vehicle at the fuel stop.

(20) The real-time driver/operator profile comprises amount of time driving during a pre-determined time period, number of rest breaks during the pre-determined time period, license compliance data, physical disabilities and driving violations.

(21) The profile of the fuel stop preferably comprises real-time types of fuels available, set billing instructions, physical dimensions of a plurality of fuel pumps, GPS coordinates, hours of operation, food service availability, and resting area availability.

(22) The configuration of the vehicle is selected from one of a single trailer, a dual trailer, a triple trailer, and a refrigeration trailer.

(23) The predetermined fueling time period is a time range to fuel the vehicle based on the real-

time GPS location of the vehicle, the real-time speed of the vehicle, the distance to the selected fuel stop from the real-time GPS location of the vehicle, and the hours of operation of the fuel stop.

(24) The dynamic compliance rules comprise speed limits, transport of toxic waste, the transport of refrigerated cargo, the rest durations for drivers/operators, the necessary insurance coverage, and the type of taxes and fees to be paid.

(25) The workflow preferably comprises an origination location of the vehicle, a destination of the vehicle, a route to the destination, a cargo, a time of departure and a time of arrival.

(26) A system **10** for securely connecting a wireless device to a single access point in a vehicle for a predetermined work assignment is set for the FIGS. **1** and **1A**. The system **10** preferably comprises a remote server (cloud) **11**, a vehicle gateway device **130**, a smart device **110** and a passive device **61**. The vehicle gateway device **130** is preferably a connected vehicle device (“CVD”).

(27) The server/cloud **11** accesses dataset **12** and obtains driver information. Vehicle information, mobile device information (MAC address), passive device information (beacon ID) and other information to compile a SCP packet **14**. At block **15**, the server **11** provides SCP definitions to the vehicle gateway device **130** and the mobile device **110**. At block **16** the server/cloud **11** authorizes the SCP. At block **17**, the server/cloud **11** communicates with the vehicle gateway device **130**.

(28) The vehicle gateway device **130** uses datasets **22**, with the beacon ID **23**, a scan of wireless devices **24** along with the SCP definitions **26** received from the server/cloud **11** to compile a CVD compiled SCP packet **25**. The CVD compiled SCP packet is sent to the cloud/server **11** at block **16** and authorization/validation of the CVD compiled SCP packet is received at block **27**. At block **28** the SCP is authorized for broadcasting at the vehicle gateway device **130** a wireless network with a hidden and hashed SSID unique to the vehicle, the hidden and hashed SSID generated from the validated SCP packet. At block **29**, the vehicle gateway device **130** communicates the broadcast with the server/cloud **11**. At block **31**, the vehicle gateway device **130** communicates with other devices, namely the smart device **110** over preferably a WiFi hotspot **32** and the passive device **61** by pairing using a BLUETOOTH communication protocol at block **33**.

(29) At block **49**, the smart device (mobile device) **110** compiles a complied mobile device SCP packet from the SCP definitions **42**, the data sets **48**, the beacon ID **43**, the Tablet ID **45**, a driver ID **46**, a vehicle ID **47** and scan of wireless devices **44**. The mobile device **110** generates the hashed SSID and a passphrase from the complied mobile device SCP packet. At block **51**, the mobile device **110** connects to the WiFi hotspot **32** of the vehicle device gateway **130**.

(30) The passive device **61** broadcast a unique ID at block **62** which is received by the mobile device **110** and the vehicle gateway device **130**. At block **63**, if a BLUETOOTH device, it broadcasts a BLUETOOTH advertisement at block **64**.

(31) The SCP is defined by an assigning authority in the server/cloud **11**. The server/cloud **11** sends the SCP definition and any other required data in datasets to the CVD **130** and the mobile device **110**. The CVD **130** adds the contextual data from local datasets to the sever-sent data to compile its SCP based definition. The local datasets include data wirelessly scanned from passive devices, preferably transmitting a BLUETOOTH beacon. Other local datasets include information from the vehicle. The CVD **130** sends its compiled SCP packet to the server **11** for authorization. The server **11** verifies the CVD compiled SCP packet, and if valid, the server **11** transmits a validation/approval signal to the CVD **130**. The CVD then generates an access point SSID/passphrase with SCP. Likewise, the mobile device **110** utilizes contextual data from local datasets to compile its SCP based on the definitions. The mobile device **110** connects to the access point of the CVD **130** using the SCP. The CVD **130** and the mobile device **110** also connect to the passive device **61** since it is part of the SCP definition.

(32) A predetermined work assignment is a temporal event with a fixed start and completion based on assignable boundary conditions. The assignable boundary condition is at least one of a predetermined time period, a geographical destination, and a set route. Alternatively, the assignable

boundary condition is any feature with a beginning and a termination. The assigning authority is performed by a person or persons, who have the appropriate authority and mechanisms to assign specific tasks and assets to a specific vehicle and vehicle operator or custodian, and to assign workflow assignments to the same. The predetermined work assignment is assigned to a known person or entity that has its own primary networked device accessible through a password protected user interface, a specific name and password that auto-populates or otherwise automatically satisfies a plurality of credentials requirements, wherein the plurality of credential requirements are automatically available or revoked based on the assignable boundary condition identified in a pairing event.

(33) The CVD **130** broadcasts a Wifi wireless network with a hidden and hashed SSID unique to the host vehicle and protected by a unique, dynamically generated and hashed passphrase. The vehicle ID is entered into an application on the tablet that is then converted to the same hashed SSID and passphrase, which allows the tablet to attempt to connect to the corresponding CVD Wifi network and begin communication.

(34) A method **900** for a secure connection to a wireless network of a vehicle is shown in FIG. 2. At block **901**, a server generates definitions for a SCP packet for assigning authority for a vehicle. At block **902** the server transmits the definitions for the SCP packet to a CVD and a mobile device. At block **903**, the CVD compiles the SCP packet to generate a CVD compiled SCP. At block **904**, the CVD transmits the CVD compiled SCP to the server for authorization. At block **905**, the server transmits authorization for the CVD compiled SCP from to the CVD for creation of a validated SCP. At block **906**, the mobile device generates a dataset to compile a mobile device compiled SCP. At block **907**, the CVD broadcasts at a wireless network with a hidden and hashed SSID unique to the vehicle. The hidden and hashed SSID is generated from the validated SCP packet. At block **908**, the mobile device generates the hashed SSID and a passphrase from the dataset, which allows the mobile device connect to the wireless network. At block **909**, the mobile device searches for a vehicle having the CVD broadcasting the wireless network in a hidden mode. At block **910**, the mobile device securely connects with the CVD.

(35) One embodiment utilizes a system for vehicle to mobile device secure wireless communications. The system comprises a vehicle **210**, a CVD **130**, a mobile device **110** and a passive communication device **61**. The vehicle **210** comprises an on-board computer with a memory having a vehicle identification number (VIN), a connector plug, and a motorized engine. The CVD **130** comprises a processor, a WiFi radio, a BLUETOOTH radio, a memory, and a connector for mating with the connector plug of the vehicle. The mobile device **110** comprises a graphical user interface, a mobile application, a processor, a WiFi radio, and a cellular network interface. The passive communication device **61** operates on a BLUETOOTH communication protocol. The server **11** is configured to generate a plurality of definitions for a SCP packet for assigning authority for the vehicle. The server **11** is configured to transmit the plurality of definitions for the SCP packet from the server to the CVD **130** and the mobile device **110**. The CVD **130** is configured to compile the SCP packet to generate a CVD compiled SCP. The CVD **130** is configured to transmit the CVD compiled SCP to the server **11** for authorization. The server **11** is configured to transmit authorization for the CVD compiled SCP to the CVD **130** for creation of a validated SCP. The mobile device **110** is configured to generating a dataset to compile a mobile device compiled SCP. The CVD **130** is configured to broadcast a wireless network with a hidden and hashed SSID unique to the vehicle, the hidden and hashed SSID generated from the validated SCP packet. The mobile device **110** is configured to generate the hashed SSID and a passphrase from the dataset, which allows the mobile device connect to the wireless network. The mobile device **110** is configured to search for a vehicle having the CVD broadcasting the wireless network in a hidden mode. The mobile device **110** is configured to connect to the CVD **130** over the wireless network.

(36) The dataset preferably comprises at least one of a plurality of definitions for the SCP packet, a

tablet ID, a driver ID, a vehicle ID, a beacon ID, identified or defined entity/participant to the transaction, descriptions, actions, or states of thing, characteristics of identifiable devices, when present in a certain proximity and/or context.

(37) Optionally, the mobile device **110** connects to a passive device, the passive device operating on a BLUETOOTH communication protocol. The passive device **61** is preferably a BLUETOOTH enabled device advertising a unique ID as a beacon or a complex system (speaker, computer, etc.) that emits BLUETOOTH enabled device advertising a unique ID as a beacon.

(38) The mobile device **110** preferably receives input from a driver of the vehicle, and/or the server **11** contains the assigning authority that generates the SCP definitions.

(39) The passive device **61** is preferably an internal device in the vehicle or an external device posted on a gate to a facility and generating a beacon. The beacon from the passive device is preferably a mechanism to ensure that the connection between the mobile device **110** and the CVD **130** occurs at a specific physical location dictated by the assigning authority through the server **11**. Preferably, the automatic connection between the mobile device **110** and the CVD occurs because the assigning authority, through the server, has dictated that it occur.

(40) As shown in FIG. 3, each of a multitude of trucks **210a-210d** broadcast a wireless signal for a truck specific network, with one truck **210c** broadcasting a wireless signal **225**. However, the SSID is not published so unless a driver is already in possession of the SSID, the driver will not be able to pair the tablet computer **110** with the CVD **130** of the truck **210** to which the driver is assigned. So even though the wireless signals are being “broadcast”, they will not appear on a driver's tablet computer **110** (or other mobile device) unless the tablet computer **110** has already been paired with the CVD **130** of the vehicle **210**. A driver **205** in possession of a tablet computer **110** pairs, using a signal **230**, the tablet computer **110** with the wireless network **225** of the CVD of the truck **210c**, and thus the driver locates the specific truck **210c** he is assigned to in a parking lot full of identical looking trucks **210a-d**.

(41) For example, on an IPHONE® device from Apple, Inc., the “UDID,” or Unique Device Identifier is a combination of forty numbers and letters, and is set by Apple and stays with the device forever.

(42) For example, on an ANDROID based system, one that uses Google Inc.'s ANDROID operating system, the ID is set by Google and created when an end-user first boots up the device. The ID remains the same unless the user does a “factory reset” of the phone, which deletes the phone's data and settings.

(43) The mobile communication device **110**, or mobile device, is preferably selected from mobile phones, smartphones, tablet computers, PDAs and the like. Examples of smartphones and the device vendors include the IPHONE® smartphone from Apple, Inc., the DROID® smartphone from Motorola Mobility Inc., GALAXY S® smartphones from Samsung Electronics Co., Ltd., and many more. Examples of tablet computing devices include the IPAD® tablet computer from Apple Inc., and the XOOM™ tablet computer from Motorola Mobility Inc.

(44) The mobile communication device **110** then a communication network utilized preferably originates from a mobile communication service provider (aka phone carrier) of the customer such as VERIZON, AT&T, SPRINT, T-MOBILE, and the like mobile communication service providers, provide the communication network for communication to the mobile communication device of the end user.

(45) Wireless standards utilized include 802.11a, 802.11b, 802.11g, AX.25, 3G, CDPD, CDMA, GSM, GPRS, radio, microwave, laser, Bluetooth, 802.15, 802.16, and IrDA.

(46) BLUETOOTH™ technology operates in the unlicensed 2.4 GHz band of the radio-frequency spectrum, and in a preferred embodiment the secondary device **30** and/or primary device **25** is capable of receiving and transmitting signals using BLUETOOTH™ technology. LTE Frequency Bands include 698-798 MHz (Band 12, 13, 14, 17); 791-960 MHz (Band 5, 6, 8, 18, 19, 20); 1710-2170 MHz (Band 1, 2, 3, 4, 9, 10, 23, 25, 33, 34, 35, 36, 37, 39); 1427-1660.5 MH (Band 11, 21,

24); 2300-2700 MHz (Band 7, 38, 40, 41); 3400-3800 MHz (Band 22, 42, 43), and in a preferred embodiment the secondary device **30** and/or the primary device **25** is capable of receiving and transmitting signals using one or more of the LTE frequency bands. WiFi preferably operates using 802.11a, 802.11b, 802.11g, 802.11n communication formats as set for the by the IEEE, and in a preferred embodiment the secondary device **30** and/or the primary device **25** is capable of receiving and transmitting signals using one or more of the 802.11 communication formats. Near-field communications (NFC) may also be utilized.

(47) As shown in FIG. 4, a typical mobile communication device **110** preferably includes an accelerometer **301**, I/O (input/output) **302**, a microphone **303**, a speaker **304**, a GPS chipset **305**, a Bluetooth component **306**, a Wi-Fi component **307**, a 3G/4G component **308**, RAM memory **309**, a main processor **310**, an OS (operating system) **311**, applications/software **312**, a Flash memory **313**, SIM card **314**, LCD display **315**, a camera **316**, a power management circuit **317**, a battery **318** or power source, a magnetometer **319**, and a gyroscope **320**.

(48) Each of the interface descriptions preferably discloses use of at least one communication protocol to establish handshaking or bi-directional communications. These protocols preferably include but are not limited to XML, HTTP, TCP/IP, Serial, UDP, FTP, Web Services, WAP, SMTP, SMPP, DTS, Stored Procedures, Import/Export, Global Positioning Triangulation, IM, SMS, MMS, GPRS and Flash. Databases that may be used with the system preferably include but are not limited to MSSQL, Access, MySQL, Progress, Oracle, DB2, Open Source DBs and others. Operating system used with the system preferably include Microsoft 2010, XP, Vista, 2000 Server, 2003 Server, 2008 Server, Windows Mobile, Linux, Android, Unix, I series, AS 400 and Apple OS.

(49) The underlying protocol at the cloud server **11**, is preferably Internet Protocol Suite (Transfer Control Protocol/Internet Protocol ("TCP/IP")), and the transmission protocol to receive a file is preferably a file transfer protocol ("FTP"), Hypertext Transfer Protocol ("HTTP"), Secure Hypertext Transfer Protocol ("HTTPS") or other similar protocols. The transmission protocol ranges from SIP to MGCP to FTP and beyond. The protocol at the authentication server **40** is most preferably HTTPS.

(50) Wireless standards include 802.11a, 802.11b, 802.11g, AX.25, 3G, CDPD, CDMA, GSM, GPRS, radio, microwave, laser, Bluetooth, 802.15, 802.16, and IrDA.

(51) Components of a cloud computing server **40** of the system, as shown in FIG. 5, preferably includes a CPU component **401**, a graphics component **402**, PCI/PCI Express **403**, memory **404**, non-removable storage **407**, removable storage **408**, Network Interface **409**, including one or more connections to a fixed network, and SQL database(s) **45a-45d**, which includes the venue's CRM. Included in the memory **404**, is an operating system **405**, a SQL server **406** or other database engine, and computer programs/software **410**. The server **40** also preferably includes at least one computer program configured to receive data uploads and store the data uploads in the SQL database. Alternatively, the SQL server can be installed in a separate server from the server **40**.

(52) A flow chart for an alternative method **600** for a secure connection to a wireless network of a vehicle is shown in FIG. 6. At block **601**, the CVD broadcasts an encrypted, blind SSID based on specific vehicle data. At block **602**, leveraging the known vehicle data and the encryption algorithm a mobile device searches for a vehicle having a CVD broadcasting the wireless network. At block **603**, the mobile device is connected with the CVD.

(53) A system for a secure connection to a wireless network of a vehicle is shown in FIG. 7. A truck **210a**. Those skilled in the pertinent art will recognize that the truck **210a** may be replaced by any type of vehicle (such as a bus, sedan, pick-up, sport utility vehicle, limousine, sports car, delivery truck, van, mini-van, motorcycle, and the like) without departing from the scope of spirit of the present invention. The truck **210a** preferably comprises a motorized engine **234**, a vehicle identification number ("VIN"), an on-board computer **232** with a memory **231** and a connector plug **235**. The on-board computer **232** preferably has a digital copy of the VIN in the memory **231**. The on-board computer **232** is preferably in communication with the motorized engine **234**. The

truck **210a** may also have a GPS component for location and navigation purposes, a satellite radio such as SIRIUS satellite radio, a driver graphical interface display, a battery, a source of fuel and other components found in a conventional long distance truck.

(54) Also in the truck **210a** is a CVD **130** comprising a processor, a WiFi radio, a BLUETOOTH radio, a memory and a connector to connect to the connector plug of the on-board computer **232**.

(55) A driver **205** preferably has a mobile communication device such as a tablet computer **110** in order to pair with a wireless network generated by the CVD **130** of the truck **210a**. The tablet computer **110** preferably comprises a graphical user interface **335**, a processor **310**, a WiFi radio **307**, a BLUETOOTH radio **306**, and a cellular network interface **308**.

(56) As shown in FIG. **8**, each of a multitude of trucks **210a-210k** broadcast a wireless signal **224a-k** for a truck specific network, with one truck **210f** broadcasting a wireless signal **225**. However, all of the wireless signal **224a-224k** and **225** do not publish their respective SSID so that a mobile device **110** must already be paired with the CVD **130** of the truck **210** in order to connect to the truck based wireless network **224a-224k** or **225** of each of the CVDs **130** of each of the trucks **210a-210k**. A driver **205** in possession of a tablet computer **110** pairs with the specific truck wireless network **225** of the CVD **130** of the truck **210f**, and thus the driver locates the specific truck **210f** he is assigned to in a parking lot full of identical looking trucks **210a-210k**.

(57) FIG. **9** is a block diagram of a system **920** for instructing a vehicle **925** where and when to refuel. The vehicle **925** begins at an origination location **901**. The vehicle **925** will travel on a route **922** to a destination location **950**. Along the route are multiple fuels stops **930a-930e**. A GPS satellite **926** provides GPS location information to the vehicle **925**. The vehicle **925** is in communication with a server **923** over a communication network **924**. The server **923** receives a workflow for a vehicle **925**. The workflow comprises an origination location of the vehicle, a destination of the vehicle, a route to the destination, a cargo, a time of departure and a time of arrival. The server **923** calculates a fuel stop for the vehicle **925** based on the workflow, the real-time operator profile, the configuration of the vehicle **925**, the real-time GPS location of the vehicle **925** (received from the vehicle), the real-time vehicle data, a dynamic compliance rule, and the selected fuel station profile. The server **923** transmits to the vehicle **925** over the communication network **924** the guidance to the selected fuel stop from the current location of the vehicle **925**. The guidance includes micro-navigation to an exact fuel pump contained within the fuel stop location of the fuel stop for fueling during a predetermined fueling time period. The dynamic compliance rules include speed limits, transport of toxic waste, the transport of refrigerated cargo, the rest durations for drivers, the necessary insurance coverage, the type of taxes and fees to be paid, and the like. The server **923** retrieves real-time compliance rules for the location of the truck from the plurality of databases, which are preferably State vehicle databases, municipal vehicle databases, county vehicle databases, and Federal vehicle databases.

(58) FIG. **10** is an illustration of multiple sensors on a truck **1000**. The vehicle/truck **1000** preferably comprises an oil level sensor **1005**, an engine sensor **1010**, a power sensor **1015**, a refrigeration/HVAC sensor **1020**, a temperature sensor **1025**, a tire pressure sensor **1030**, and a fuel sensor **1035**. Those skilled in the pertinent art will recognize that multiple other sensors may be utilized without departing from the scope and spirit of the present invention.

(59) FIG. **11** is a flow chart for a method **1100** for instructing a vehicle where and when to refuel. At block **1101**, a server receives a workflow for a vehicle. The workflow comprises an origination location of the vehicle, a destination of the vehicle, a route to the destination, a cargo, a time of departure and a time of arrival. At block **1102**, the server receives a real-time operator profile for the operator of the vehicle. At block **1103**, the server receives real-time data for the vehicle from a CVD connected to on board diagnostics for the vehicle. At block **1104**, the server receives a configuration of the vehicle. At block **1105**, the server receives at least one dynamic compliance rule, and preferably multiple dynamic compliance rules. At block **1106**, the server receives at least one configurable real-time vehicle data trigger event. At block **1107**, the server utilizes a real-time

GPS location for the vehicle. The GPS location is preferably received from the vehicle. At block **1108**, the server determines at least one fuel stop along the route, and most preferably multiple fuel stops. At block **1109**, the server calculates a fuel stop for the vehicle based on the workflow, the real-time operator profile, the configuration of the vehicle, the real-time GPS location of the vehicle, the real-time vehicle data, dynamic compliance rule, and the selected fuel station profile. At block **1110**, the server transmits to the vehicle guidance to the selected fuel stop from the current location of the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump contained within the fuel stop location of the fuel stop for fueling during a predetermined fueling time period.

(60) FIG. **12** is a flow chart for a method **1200** for instructing a vehicle where and when to refuel. At block **1201**, a real-time GPS location for a vehicle is determined. At block **1202**, at least one fuel stop along a route for the vehicle is determined, and preferably multiple fuel stops are determined. At block **1203**, a fuel stop is calculated for the vehicle based on a workflow, a real-time operator profile, a configuration of the vehicle, the real-time GPS location of the vehicle, real-time vehicle data, at least one dynamic compliance rule, and a selected fuel stop profile. At block **1204**, guidance to the selected fuel stop from the current location of the vehicle is transmitted to the vehicle, wherein the guidance includes micro-navigation to an exact fuel pump contained within the fuel stop location of the fuel stop for fueling during a predetermined time period.

(61) Kennedy, et al., U.S. patent application Ser. No. 16/450,959, filed on Jun. 24, 2019 for Secure Wireless Networks For Vehicles, is hereby incorporated by reference in its entirety.

(62) Son et al., U.S. Pat. No. 10,475,258 for a Method And System For Utilizing Vehicle Odometer Values And Dynamic Compliance is hereby incorporated by reference in its entirety.

(63) From the foregoing it is believed that those skilled in the pertinent art will recognize the meritorious advancement of this invention and will readily understand that while the present invention has been described in association with a preferred embodiment thereof, and other embodiments illustrated in the accompanying drawings, numerous changes modification and substitutions of equivalents may be made therein without departing from the spirit and scope of this invention which is intended to be unlimited by the foregoing except as may appear in the following appended claim. Therefore, the embodiments of the invention in which an exclusive property or privilege is claimed are defined in the following appended claims.

Claims

1. A method for providing driving instructions to a vehicle, the method comprising: receiving at a server a workflow data for a vehicle, the workflow data comprising an origination location of the vehicle, a destination of the vehicle, a cargo, and a time of departure; utilizing at the server a real-time location for the vehicle; determining at the server at least one stop along a route from the origination location to the destination of the vehicle where the vehicle can replenish; calculating the at least one fuel stop for the vehicle based on the workflow, the real-time location of the vehicle, a plurality of power unit values, the workflow data, and a location of the at least one stop; and transmitting to the vehicle guidance to the at least one stop from the real-time location of the vehicle, wherein the guidance includes micro-navigation within the stop location for vehicle replenishment during a predetermined time period.

2. The method according to claim 1 further comprising receiving at the server a real-time speed of the vehicle.

3. The method according to claim 1 further comprising receiving at the server at least one configurable real-time vehicle data trigger event.

4. The method according to claim 1 further comprising automatically billing an off-site entity for the fuel pumped by the vehicle at the fuel stop.

5. The method according to claim 1 further comprising receiving at the server a real-time operator

profile comprising at least one of amount of time driving during a pre-determined time period, number of rest breaks during the pre-determined time period, license compliance data, physical disabilities or driving violations.

6. The method according to claim 1 further comprising receiving at the server a profile of the stop comprising at least one of physical layouts and barriers, obstacles or other limiting constraints that could potentially impede the transit of specific vehicles based on their physical characteristics, GPS coordinates of the stop, hours of operation, food service availability, or resting area availability.

7. The method according to claim 1 further comprising receiving at the server a configuration of the vehicle selected from one of a single vehicle, a vehicle with one or more trailers, and a vehicle with one or more refrigeration trailers.

8. The method according to claim 1 wherein the predetermined time period estimate is a time range describing the expected arrival time in GMT at the stop based on the real-time GPS location of the vehicle, the real-time speed of the vehicle, the distance to the selected stop from the real-time GPS location of the vehicle, and the hours of operation of the stop.

9. The method according to claim 1 further comprising receiving at the server a dynamic compliance rule comprising at least one of speed limits, transport of toxic waste, the transport of refrigerated cargo, the rest durations for operators, the necessary insurance coverage, or the type of taxes and fees to be paid.

10. A method for providing driving instructions to a vehicle, the method comprising: receiving, at a server, a real-time GPS location for a vehicle; determining, at a server, at least one stop along a route for the vehicle to replenish; calculating, at the server, driving directions from the real-time GPS location for the vehicle to the at least one stop for the vehicle based on a workflow, a real-time operator profile, a real-time location of the vehicle, at least one power unit value, a real-time vehicle data, at least one dynamic compliance rule, and a selected stop profile; and transmitting from the server to the vehicle driving directions to the selected stop from the real-time GPS location of the vehicle, wherein the driving instructions include micro-navigation for replenishing the vehicle during a predetermined time period.

11. The method according to claim 10 wherein the workflow comprises an origination location of the vehicle, a destination of the vehicle, a route to the destination, a cargo, a time of departure or a time of arrival.

12. The method according to claim 10 wherein the real-time operator profile comprises amount of time driving during a pre-determined time period, number of rest breaks during the pre-determined time period, license compliance data, physical disabilities and driving violations.

13. The method according to claim 10 wherein the predetermined time period is a time range to replenish the vehicle based on the real-time GPS location of the vehicle, the real-time speed of the vehicle, the distance to the selected stop from the real-time GPS location of the vehicle, and the hours of operation of the stop.
